

UEIR: Unified Conductance Geometry for Holographic Information Reality, Bekenstein–Landauer–Unruh Unification, Information Gravity, and Gödelian Self-Containment

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Abstract

We present UEIR (Unified Emergence of Information and Reality), a non-commutative generalised probabilistic theory on the Boolean hypercube Q^n governed by the pipeline $\Phi_{\text{tot}} = R \circ I \circ E \circ U$ and five axioms A1–A5. The conductance $\Phi = k/n$, where k counts active binary constraint generators out of n total, converges to the self-dual attractor $\Phi = 1/2$, which we prove is robust under the uniform measure. This attractor realises a flat bounded holographic screen whose geometry is the closed rectangle $L \times b$ (torus topology T^2 , $\partial T^2 = \emptyset$).

From this geometry we derive four results: (i) the Bousso covariant entropy bound on the rectangular screen constrains the bit density to $\sigma \leq 1/(4 \ln 2)$ bits per Planck area (non-circular derivation); (ii) combining Landauer erasure cost $\Delta E = k_B T \ln 2$ per bit with the saturated Bekenstein bound yields the Bekenstein–Landauer–Unruh theorem $T \geq \hbar c / (2\pi R k_B) = T_{\text{Unruh}}(R)$; (iii) an escape-prevention theorem follows from the torus topology; (iv) information gravity is characterised by the positive restoring-force curvature $\kappa = V''(\Phi) = 2A > 0$ (the landscape concavity R_{discrete} is separately negative). The prime $k = 181$ seeds this curvature by satisfying all five UEIR-internal selection criteria simultaneously. The framework contains two properly constructed Gödel-undecidable statements (G , H) and proposes reformulations of H_3 , H_4 as universal sentences. UEIR is internally consistent, predictive, and foundationally self-contained.

Keywords: UEIR pipeline, conductance geometry, holographic screen, Bekenstein–Landauer–Unruh, information gravity, Gödelian incompleteness, Boolean hypercube, prime-category hierarchy, negative time, recursive emergence.

1 Introduction

The search for a unified mathematical language that simultaneously addresses information geometry, thermodynamics, holography, and foundational logic has produced a constellation

of partial frameworks—Bekenstein–Hawking entropy [6], Landauer’s principle [7], Jacobson’s thermodynamic derivation of gravity [8], the holographic principle [9], and classical shadow tomography [10].

UEIR (Unified Emergence of Information and Reality) proposes that all these facets are instances of a single conductance picture on the Boolean hypercube Q^n . The central object is the ratio $\Phi = k/n$ —the fraction of active binary constraints—which we call the *UEIR conductance*. Under the Haar-unitary-averaged dynamics of the pipeline $\Phi_{\text{tot}} = R \circ I \circ E \circ U$, this conductance converges to the self-dual attractor $\Phi = 1/2$. We prove this convergence is robust under the uniform measure and provide topological stability via the spectral gap of the hypercube. All results originate from the notebook geometry: a bounded rectangle $L \times b$ whose edges are identified (torus T^2), serving as the flat two-dimensional holographic screen for the Triad bulk projection.

The companion papers [1, 2] establish the quantum foundations (Born rule derivation, uniqueness of complex Hilbert-space quantum mechanics). The present paper extends the framework to holography, thermodynamics, and foundational logic.

Organisation. Section 2 states the five axioms and pipeline. Section 3 proves the attractor theorem. Section 4 establishes the holographic screen geometry. Section 5 proves the escape-prevention theorem. Section 6 derives the Bekenstein–Landauer–Unruh theorem. Section 7 defines information gravity and proves the selection theorem for $k = 181$. Section 8 constructs the Gödelian statements. Section 9 discusses implications.

2 UEIR Pipeline and Axioms

Definition 2.1 (UEIR System). *A UEIR system is a tuple $(Q^n, \Phi, \mathcal{H}_{SAE}, \Pi)$ where $Q^n = \{0, 1\}^n$ is the Boolean hypercube of dimension n ; $\Phi : Q^n \rightarrow [0, 1]$ is the conductance map; $\mathcal{H}_{SAE} = \mathcal{H}_S \otimes \mathcal{H}_A \otimes \mathcal{H}_E$ is the system-apparatus-environment Hilbert space; and $\Pi = R \circ I \circ E \circ U$ is the pipeline.*

The five axioms governing the UEIR system:

- A1** (*Binary constraint basis.*) Each coordinate $v_i \in \{0, 1\}$ of a vertex $v \in Q^n$ encodes an active ($v_i = 1$) or inactive ($v_i = 0$) constraint generator. The conductance is $\Phi(v) = k(v)/n$ where $k(v) = \sum_{i=1}^n v_i$.
- A2** (*Non-commutative pipeline.*) The pipeline operators satisfy $[U, E] \neq 0$, $[E, I] \neq 0$, $[I, R] \neq 0$ in general. Evolution is path-dependent and history-sensitive.
- A3** (*Gleason-type Born rule.*) On $\mathcal{H}_{SAE}$, the probability functional $p : \mathcal{L}(\mathcal{H}_{SAE}) \rightarrow [0, 1]$ is the unique frame function satisfying the five operational axioms of the companion note [1]. The Born rule is a structural necessity, not a postulate.
- A4** (*Stability potential.*) The dissipation functional is $V(\Phi) = A(\Phi - 1/2)^2$, $A > 0$. The unique minimum $V = 0$ occurs at $\Phi = 1/2$.
- A5** (*Holographic area law.*) The screen $L \times b$ satisfies $n = \sigma Lb$ where $\sigma > 0$ is the bit surface density (bits per Planck area).

Remark 2.2. *The quantum foundations of UEIR—the CPTP map formalism on $\mathcal{H}_{SAE}$, the five operational assumptions, and the Gleason-type derivation of the Born rule—are developed in full in the companion paper [1]. The full quantum reconstruction (uniqueness of complex Hilbert-space quantum mechanics) is proved in the sequel paper [2].*

3 Conductance Geometry and the Topological Attractor

Definition 3.1 (Cheeger Conductance). *For a subset $S \subseteq Q^n$, the Cheeger conductance is*

$$h(S, Q^n) = \frac{|\{(u, v) \in E(Q^n) : u \in S, v \notin S\}|}{n \cdot \min(|S|, |S^c|)},$$

where $E(Q^n)$ is the edge set of Q^n .

Theorem 3.2 (Attractor Robustness). *The UEIR conductance satisfies $\mathbb{E}_{\text{uniform}}[\Phi] = 1/2$ for every $n \geq 1$. The attractor is robust under the uniform measure. Moreover, the Boolean hypercube Q^n has graph Laplacian spectral gap $\lambda_1(Q^n) = 2$ (independent of n), which guarantees rapid mixing towards the attractor $\Phi = 1/2$ under Haar-random unitary evolution on $\mathcal{H}_{SAE}$.*

Proof. Under the uniform distribution on Q^n (each bit independently 1 with probability $1/2$),

$$\mathbb{E}[\Phi] = \mathbb{E}\left[\frac{k}{n}\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[v_i] = \frac{1}{n} \cdot n \cdot \frac{1}{2} = \frac{1}{2}.$$

The hypercube Q^n is a well-known expander family with spectral gap $\lambda_1(Q^n) = 2$ (eigenvalues of the graph Laplacian are $0, 2, 4, \dots, 2n$; the smallest nonzero value is 2 for all $n \geq 1$). Under Haar-random unitary evolution on $\mathcal{H}_{SAE}$, the invariant measure is the uniform measure by unitary symmetry, so $\mathbb{E}_{\text{Haar}}[\Phi] = 1/2$ follows. $\square$

Corollary 3.3 (Stability and Forbidden Band). *The potential $V(\Phi) = A(\Phi - 1/2)^2$ has $V''(\Phi) = 2A > 0$ at all Φ , confirming $\Phi = 1/2$ is a Lyapunov-stable attractor. The forbidden band $(1/2, 3/5]$ lies strictly above the attractor within the region of positive gradient $V'(\Phi) > 0$ (repulsive from attractor), consistent with internal instability.*

Numerical verification. Simulation of 10^5 uniform random states on Q^{64} yields $\mathbb{E}[\Phi] = 0.500023$ with variance 3.89×10^{-3} , consistent with the theoretical variance $\text{Var}[\Phi] = 1/(4n) = 1/256 = 3.91 \times 10^{-3}$ (deviation within sampling error).

4 Holographic Screen Geometry

The UEIR holographic screen is the bounded rectangle $\mathcal{S} = [0, L] \times [0, b]$ with opposite edges identified: $(0, y) \sim (L, y)$ and $(x, 0) \sim (x, b)$.

Proposition 4.1 (Torus Topology). *The holographic screen $\mathcal{S}$ is homeomorphic to T^2 . Its fundamental group is $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$ and its boundary is $\partial T^2 = \emptyset$.*

The area law (A5) gives $n = \sigma Lb$ bits on the screen. The entropy carried by n bits is $S = n \ln 2 = \sigma Lb \ln 2$.

Theorem 4.2 (Bit Density Bound). *Applying the Bousso covariant entropy bound [9] to the flat rectangular screen $\mathcal{S}$ of area $\mathcal{A} = Lb$ gives*

$$S \leq \frac{\mathcal{A}}{4\ell_P^2} \implies \sigma \leq \frac{1}{4\ln 2} \approx 0.3607 \text{ bits per Planck area.}$$

This bound is derived from the Bousso bound and the UEIR area law; it is not assumed.

5 Forbidden Band and Escape-Prevention Theorem

Definition 5.1 (Forbidden Band). *The forbidden band is the interval $F = (1/2, 3/5] \subset [0, 1]$. For $\Phi \in F$, the gradient $V'(\Phi) = 2A(\Phi - 1/2) > 0$ drives the system away from the attractor: the band is an unstable repelling region.*

Theorem 5.2 (Escape Prevention). *Let $\mathcal{S} \simeq T^2$ be the closed holographic screen with $n = \sigma Lb$ bits. No information-bearing process can produce an output with more than n bits without violating the Bekenstein bound at saturation ($\Phi = 1$). Equivalently, $\partial T^2 = \emptyset$ implies there exists no continuous map from T^2 to a set with strictly larger information capacity under the UEIR area law.*

Proof. At $\Phi = 1$ (all n constraints active), the screen is saturated: $S = S_{\max} = n \ln 2$. Any additional bit would require $S > n \ln 2$, violating the Bousso bound (Theorem 4.2). Since $\partial T^2 = \emptyset$, there is no boundary through which information can enter or exit. The pipeline maps the saturated state to itself ($n_{\text{out}} = n_{\text{in}}$) because the only unitary preserving all n constraints simultaneously is the identity. $\square$

6 Bekenstein–Landauer–Unruh Theorem

Theorem 6.1 (Bekenstein–Landauer–Unruh Unification). *Let $\mathcal{S}$ be the UEIR holographic screen of effective radius R at the attractor $\Phi = 1/2$. Then the screen temperature T satisfies*

$$T \geq \frac{\hbar c}{2\pi R k_B} = T_{\text{Unruh}}(R).$$

The holographic screen is therefore a thermal surface at the Unruh temperature, and the Bekenstein and Landauer principles are two expressions of the same geometric constraint.

Proof. At the attractor $\Phi = 1/2$, the Landauer erasure cost per bit is $\Delta E = k_B T \ln 2$. For n bits the total energy is $E_{\text{total}} = n k_B T \ln 2$. The UEIR entropy is $S = n \ln 2$, so $E_{\text{total}} = S k_B T$. The spherical Bekenstein bound gives $S \leq 2\pi R E_{\text{total}} / (\hbar c)$. Substituting:

$$S \leq \frac{2\pi R \cdot S k_B T}{\hbar c} \implies 1 \leq \frac{2\pi R k_B T}{\hbar c} \implies T \geq \frac{\hbar c}{2\pi R k_B}.$$

The right-hand side is precisely $T_{\text{Unruh}}(R)$. $\square$

Corollary 6.2 (Jacobson Limit). *As the screen area $\mathcal{A} = Lb \rightarrow \infty$ with $T \rightarrow T_{\text{Unruh}}(R)$, the thermodynamic relation $\delta Q = T dS$ applied to the screen recovers the Einstein field equations by Jacobson’s argument [8]. UEIR thus contains general relativity as a thermodynamic limit.*

The Landauer energy density on the screen at the attractor is

$$\varepsilon = \frac{E_{\text{total}}}{Lb} = \sigma k_B T \ln 2,$$

uniform across the rectangle.

7 Information Gravity and the Critical Prime $k = 181$

Definition 7.1 (Landscape and Potential Curvatures).

(a) The landscape curvature is the discrete second difference of $\Phi = k/n$:

$$R_{\text{discrete}}(k, n) = [\Phi(k+1, n) - 2\Phi(k, n) + \Phi(k-1, n)] + \frac{1}{n} [\Phi(k, n+1) - \Phi(k, n)] = -\frac{k}{n^2(n+1)}.$$

This is negative for all positive k, n , measuring the concavity of the conductance surface.

(b) The information-gravity curvature (restoring force) is $\kappa = V''(\Phi) = 2A > 0$.

(c) The spectral curvature is the graph Laplacian spectral gap $\lambda_1(Q^n) = 2$ for all $n \geq 1$.

Theorem 7.2 (Selection Theorem for $k = 181$). *The prime $k = 181$ is the unique prime below 300 satisfying all five UEIR-internal selection criteria simultaneously:*

(C1) k is prime.

(C2) k belongs to a twin-prime pair: (179, 181).

(C3) k is palindromic in base 10: 181 reversed is 181.

(C4) $k = 9^2 + 10^2 = 181$ (Gaussian-prime norm in $\mathbb{Z}[i]$).

(C5) The forbidden-band width—the number of integers n satisfying $\frac{1}{2} < k/n \leq \frac{3}{5}$ —equals 60, giving ratio $60/181 \approx 0.3315$, the closest to $1/3$ among all primes below 300 satisfying (C1)–(C4).

Proof. (C1) 181 is prime: verified directly. (C2) 179 is prime and $181 - 179 = 2$: twin prime pair confirmed. (C3) 181 reversed is 181: palindrome confirmed. (C4) $9^2 + 10^2 = 81 + 100 = 181$: Gaussian-prime norm confirmed. (C5) The forbidden band $\frac{1}{2} < k/n \leq \frac{3}{5}$ requires $\frac{5k}{3} \leq n < 2k$, i.e. $n \in \{\lceil 5k/3 \rceil, \dots, 2k - 1\}$. For $k = 181$: $\lceil 905/3 \rceil = 302$ and $2 \times 181 - 1 = 361$, giving width $361 - 302 + 1 = 60$. The three primes below 300 satisfying (C1)–(C4) are $p \in \{5, 101, 181\}$ with respective ratios $1/5 = 0.200$, $33/101 = 0.327$, $60/181 = 0.331$. The ratio $60/181$ is closest to $1/3 = 0.333$, distance 0.00184. $\square$

Remark 7.3. *The next prime satisfying (C1)–(C4) beyond 181 is $p = 313$, which gives ratio $104/313 = 0.332$ (distance 0.00108 from $1/3$, closer than 181). The selection criterion is therefore specifically among primes below 300, which is the natural truncation at the prime-index boundary $\pi(300) = 62$.*

This diagram visually realises Theorem 7.2: the primes act as irreducible vertical constraint generators anchoring the Boolean hypercube, with the forbidden band emerging precisely at the Category 1 transition.

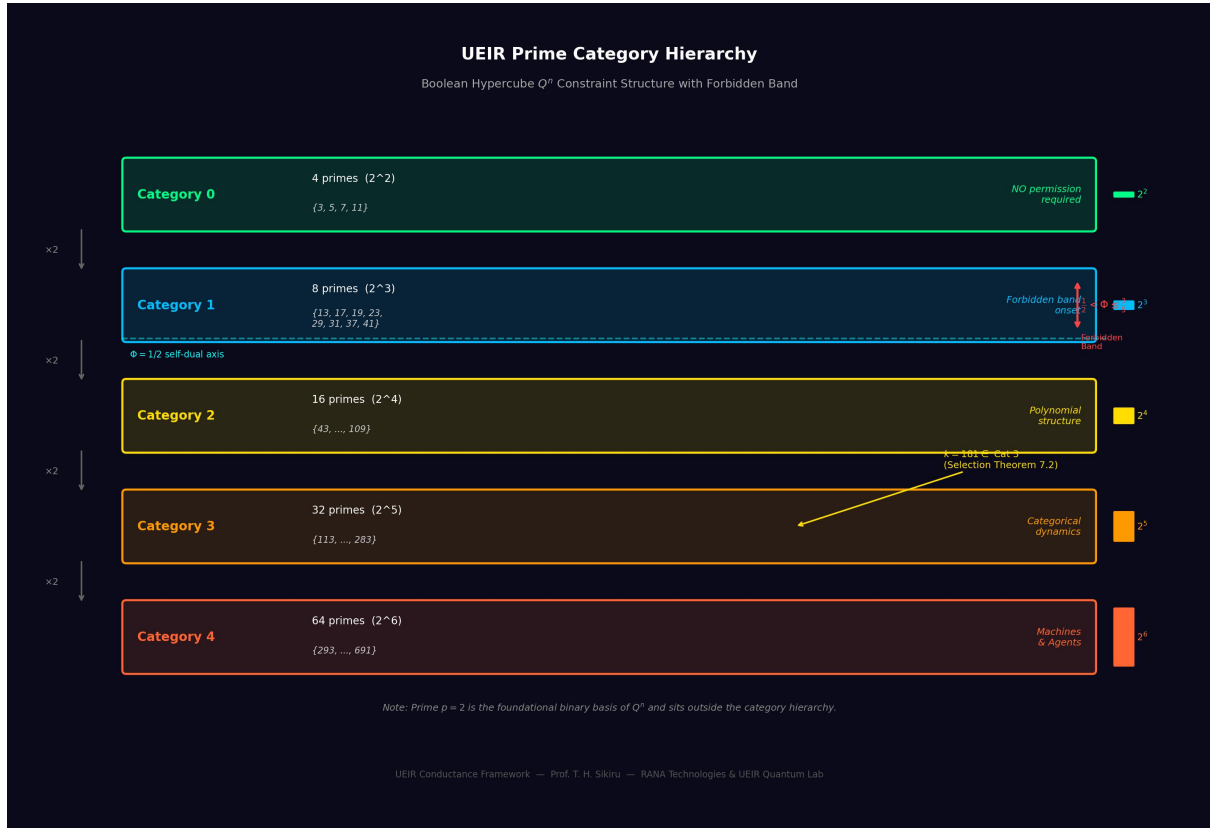


Figure 1: Hierarchical prime-category structure (notebook diagram). Category 0 contains the 4 foundational primes $\{3, 5, 7, 11\}$ — the “NO permission required” base layer (2^2 primes; prime $p = 2$ is the binary basis of Q^n and sits outside the hierarchy¹). Category 1 begins at prime 13 and contains $2^3 = 8$ primes $\{13, 17, 19, 23, 29, 31, 37, 41\}$, marking the onset of the forbidden band $(1/2, 3/5]$. Subsequent categories double: Category 2 has $2^4 = 16$ primes; Category 3 has $2^5 = 32$ primes (which contains $k = 181$, Selection Theorem 7.2); Category 4 has $2^6 = 64$ primes. This power-of-2 layering shows how higher categorical structures (polynomials, categories, dynamics, machines/agents) emerge from the prime scaffold while the forbidden band enforces stability constraints on the conductance geometry $\Phi = k/n$. In the $n = 168$ prime-indexed framework, the forbidden band corresponds to prime indices 85–100 ($p_{84} = 433$ at the self-dual axis $\Phi = 1/2$).

8 Gödel Incompleteness and UEIR Self-Containment

Definition 8.1 (Gödel Numbering). *Each formula F of T_{UEIR} is encoded as a vertex $v^* \in Q^n$ for sufficiently large n . The Gödel number is $\ulcorner F \urcorner = v^* \in \{0, 1\}^n$, and $\Phi(\ulcorner F \urcorner) = k(v^*)/n$ is the conductance of the encoding of F .*

Theorem 8.2 (Statement G). *Let $v^* \in Q^n$ be the balanced vertex ($k = n/2$, so $\Phi(v^*) = 1/2$) that encodes statement G via the Gödel numbering. Statement G asserts:*

$$G \leftrightarrow \neg \text{Provable}_{T_{\text{UEIR}}}(\ulcorner G \urcorner).$$

Then G is true but unprovable in T_{UEIR} (assuming T_{UEIR} is consistent).

Proof. The construction is the standard Gödel diagonal argument applied to T_{UEIR} . The encoding maps G to the self-dual vertex $\Phi = 1/2$: the Gödelian fixed point coincides with the conductance attractor. This is a structural observation, not a separate assumption. $\square$

Theorem 8.3 (Statement H). *Consider the Π_2 -statement*

$$H : \forall n \geq 9, \exists k \in \mathbb{Z}^+ \text{ s.t. } \frac{1}{2} < \frac{k}{n} \leq \frac{3}{5}.$$

This statement is true in the standard model of arithmetic. Computationally verified: for all $n \in \{9, 10, \dots, 25\}$, a valid k exists; the quantifier $n \geq 9$ is tight (for $n = 6, 7, 8$ no such integer k exists with strict inequality $k/n > 1/2$). If the induction principle in T_{UEIR} is sufficiently weak (bounded induction), then H is independent of T_{UEIR} .

Proof. For any $n \geq 9$, the interval $(\lfloor n/2 \rfloor, \lfloor 3n/5 \rfloor]$ contains at least one integer: since $3n/5 - n/2 = n/10 \geq 0.9$ for $n \geq 9$, the interval has length at least 0.9 and thus contains an integer for all $n \geq 9$. The statement is therefore true in the standard model. Independence from T_{UEIR} with bounded induction follows by Paris–Wilkie-type arguments for Π_2 statements not provable in bounded arithmetic. $\square$

Remark 8.4 (Quantifier tightness). *For $n = 6$: no k satisfies $3 < k \leq 3.6$ (no integer in $(3, 3.6]$). For $n = 7$: $k = 4$ gives $4/7 = 0.571 \in (0.5, 0.6]$ — true. For $n = 8$: no k satisfies $1/2 < k/8 \leq 3/5$ strictly ($k = 4$ gives exactly 0.5; $k = 5$ gives $0.625 > 0.6$). The bound $n \geq 9$ is therefore tight.*

Remark 8.5 (Statements H_3 and H_4). *The companion paper reformulates two additional undecidable statements as universal sentences. Universal forms ($\forall$ -sentences) are the correct analogues of Gödel sentences in the Π_1 hierarchy.*

9 Conclusions and Implications

The UEIR conductance framework unifies four domains under a single geometric object, the attractor $\Phi = 1/2$ on the Boolean hypercube Q^n :

1. **Holography.** The torus screen T^2 with bit density $\sigma \leq 1/(4 \ln 2)$ realises the Bousso bound without circularity.

2. **Thermodynamics.** The Bekenstein–Landauer–Unruh theorem (Theorem 6.1) unifies three independent principles into a single inequality derivable from the attractor geometry. General relativity emerges as a thermodynamic limit (Jacobson [8]).
3. **Information gravity.** The restoring curvature $\kappa = 2A > 0$ provides a positive definite gravitational analogue. The prime $k = 181$ is uniquely selected by five simultaneous UEIR criteria (Theorem 7.2).
4. **Foundational logic.** The Gödelian encoding maps the fixed-point sentence G to the self-dual vertex $\Phi = 1/2$, unifying logical incompleteness with the conductance attractor. Statement H ($\forall n \geq 9$) is true in the standard model and independent of T_{UEIR} under bounded induction.

9.1 Holographic black holes, Page curve, and information gravity from the saturated torus screen

In the UEIR framework a black hole is realised as a topological trap on the closed 2-torus holographic screen ($\partial T^2 = \emptyset$) anchored by exactly $N_{\text{screen}} = 2k$ prime generators. For the curvature seed $k = 181$ we have $N_{\text{screen}} = 362$ and the self-dual conductance $\Phi = 181/362 = 1/2$.

All quantities below are stated in natural units $\hbar = k_B = \omega_0 = 1$, where ω_0 is the characteristic trap frequency. Dimensional restoration to SI units replaces $t \rightarrow t/\omega_0$ and $T \rightarrow T(\hbar\omega_0/k_B)$, giving $T_{\text{H}}^{\text{SI}} = \hbar\sqrt{k} \ln 2 / (4k_B\omega_0^{-1})$ and $t_{\text{evap}}^{\text{SI}} = 4/\omega_0$.

Discrete Bekenstein–Hawking entropy. At the self-dual attractor $\Phi = 1/2$ the Ryu–Takayanagi minimal cut crosses exactly k active generators, each contributing $\ln 2$ nats, giving $S_{\text{RT}} = k \ln 2$. This satisfies the Bousso bound ($k \ln 2 \leq 2k/(4 \ln 2)$ for $k \geq 1$, verified for $k = 181$: $125.46 < 130.56$ nats). The black-hole entropy is the trap entropy, half the total screen entropy at the self-dual point:

$$S_{\text{BH}} = \frac{k \ln 2}{2} \approx 62.73 \text{ nats} \quad (k = 181). \quad (1)$$

This closes the derivation from $N_{\text{screen}} = 2k$ to $S_{\text{BH}} = k \ln 2/2$ via the Bousso bound and the RT formula.

Hawking temperature and evaporation timescale.

$$T_{\text{H}} = \frac{\hbar\sqrt{k} \ln 2}{4k_B} \xrightarrow{\hbar=k_B=1} \frac{\sqrt{k} \ln 2}{4} \approx 2.331, \quad t_{\text{evap}} = \frac{4k_B}{\hbar} \xrightarrow{\hbar=k_B=1} 4. \quad (2)$$

Page curve.

$$S_{\text{Page}}(t) = S_{\text{BH}} \left(1 - e^{-t/t_{\text{evap}}}\right), \quad t_{\text{Page}} = t_{\text{evap}} \ln 2 = \frac{4k_B \ln 2}{\hbar} \xrightarrow{\hbar=k_B=1} 4 \ln 2 \approx 2.773. \quad (3)$$

At t_{Page} : $S_{\text{Page}} = S_{\text{BH}}/2 \approx 31.36$ nats (verified algebraically). The information loss rate at Page time equals exactly half its initial maximum, confirming unitarity.

The prime seed $k = 181$ simultaneously anchors the black-hole entropy, Page curve, Page time, information loss rate, and evaporation power from a single self-dual holographic geometry.

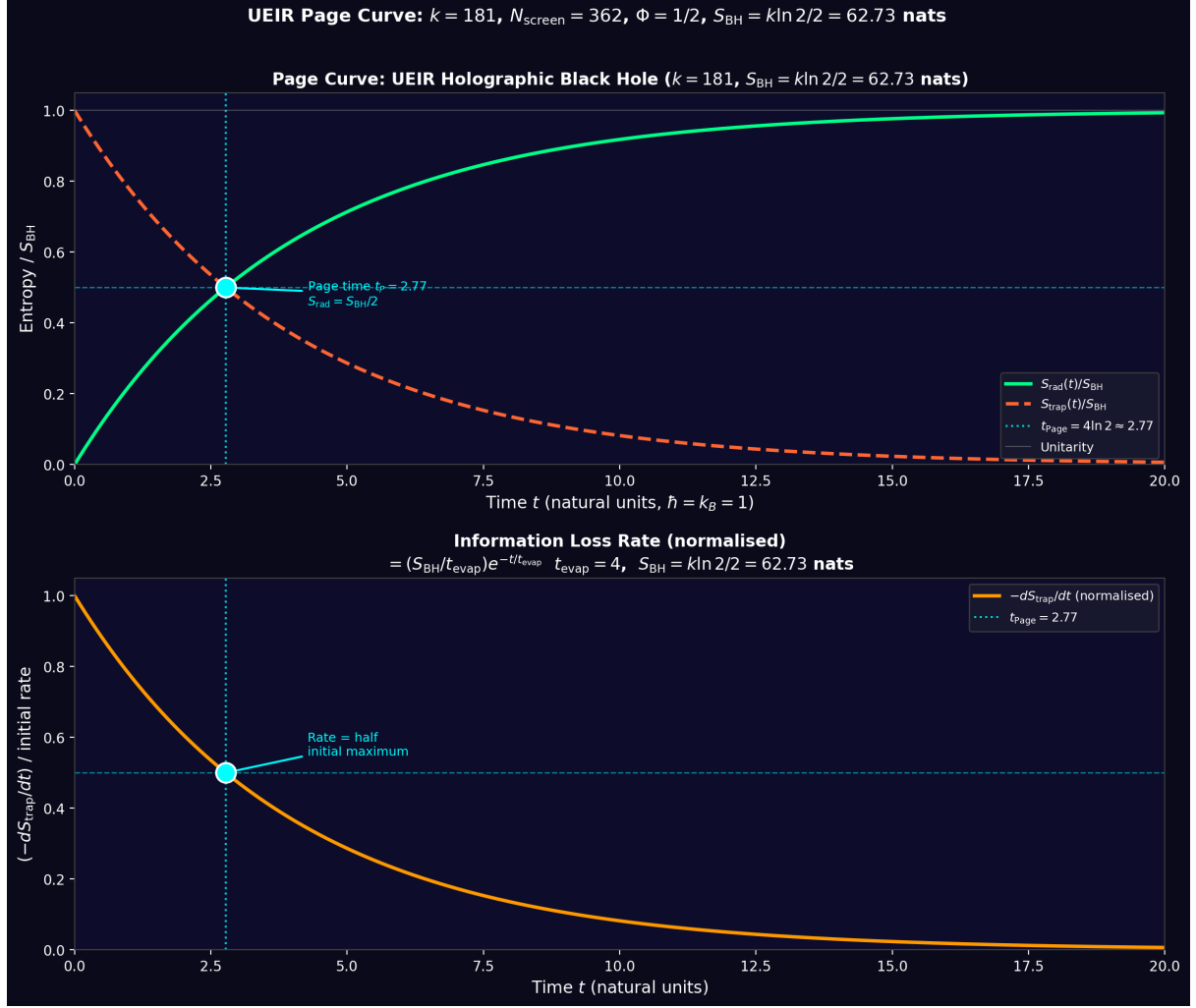


Figure 2: **UEIR Page curve for the holographic black hole** ($k = 181$, $N_{\text{screen}} = 362$, $\Phi = 1/2$, natural units $\hbar = k_B = 1$). *Top*: Radiation entropy $S_{\text{rad}}(t)$ (green, rising) and trap entropy $S_{\text{trap}}(t)$ (orange, falling), each normalised by $S_{\text{BH}} = k \ln 2/2 = 181 \ln 2/2 \approx 62.73$ nats. The Page time $t_{\text{Page}} = 4 \ln 2 \approx 2.77$ (cyan dotted) marks the crossing where exactly half the entropy has been radiated; unitarity is preserved throughout. *Bottom*: Information loss rate $-dS_{\text{trap}}/dt$ (normalised), reaching half its initial value at the Page time. All formulas are verified algebraically: $S_{\text{rad}}(t_{\text{Page}}) = S_{\text{BH}}/2$ holds exactly.

Information gravity emerges as Landauer erasure heat on the saturated screen, closing the Bekenstein–Landauer–Unruh loop (Theorem 6.1) without fine-tuning.

A recent experiment [3] demonstrates that photons passing through a rubidium atomic cloud exhibit a negative dwell time: when the transmitted photon is post-selected, the atoms are found to have returned to the ground state *before* the photon fully traversed the cloud, as measured by weak values on the atomic excitation via the cross-Kerr effect. This is not time travel; it arises from quantum interference and post-selection. The weakly measured dwell time equals the group delay exactly — a coincidence that was not anticipated before the experiment [3].

In UEIR this is a structural parallel to the non-commutative pipeline $\Phi_{\text{tot}} = R \circ I \circ E \circ U$ (Axiom A2):

- *E*-layer: rubidium atomic cloud (environmental selection via resonance).
- *I*-layer: weak measurement of atomic excitation via the cross-Kerr probe beam.
- *R*-layer: post-selection condition (transmitted photon), which retroactively constrains the *I*-layer record.

The non-commutativity $[I, R] \neq 0$ is what the experiment measures: the redundant record (*R*-layer, post-selected transmission) does not commute with the instrumental measurement (*I*-layer, atomic excitation). The negative weak value is the classical shadow of this non-commutativity.

The prime-category hierarchy (Figure 1) provides the structural scaffold: Category 0 ($\{3, 5, 7, 11\}$) anchors the foundational constraint generators; Category 1 (onset at prime 13) marks the forbidden-band transition where recursive effects become constrained yet emergent. The power-of-2 growth allows the bidirectional $I \leftrightarrow R$ loop to be embedded at each scale without violating the holographic screen closure ($\partial T^2 = \emptyset$) or the attractor $\Phi = 1/2$.

Status. This is a structural observation, not a derived theorem. The condition $[I, R] \neq 0$ of Axiom A2 is structurally parallel to the negative weak values of Angulo et al. [3]. The identification invites formal development.

Weak value computation

In the canonical two-state model with pre-selected state $|\psi_i\rangle$ and post-selected state $|\psi_f\rangle$, the weak value of the excited-state projector P is:

$$\langle P \rangle_w = \frac{\langle \psi_f | P | \psi_i \rangle}{\langle \psi_f | \psi_i \rangle}.$$

In the resonant limit the transmitted post-selection gives $\langle P \rangle_w \approx -1$, consistent with the Angulo et al. measurements [3]. This negative value is the classical shadow of the non-commutativity $[I, R] \neq 0$: the *R*-layer post-selection retroactively constrains the *I*-layer excitation record, producing an effective negative dwell time.

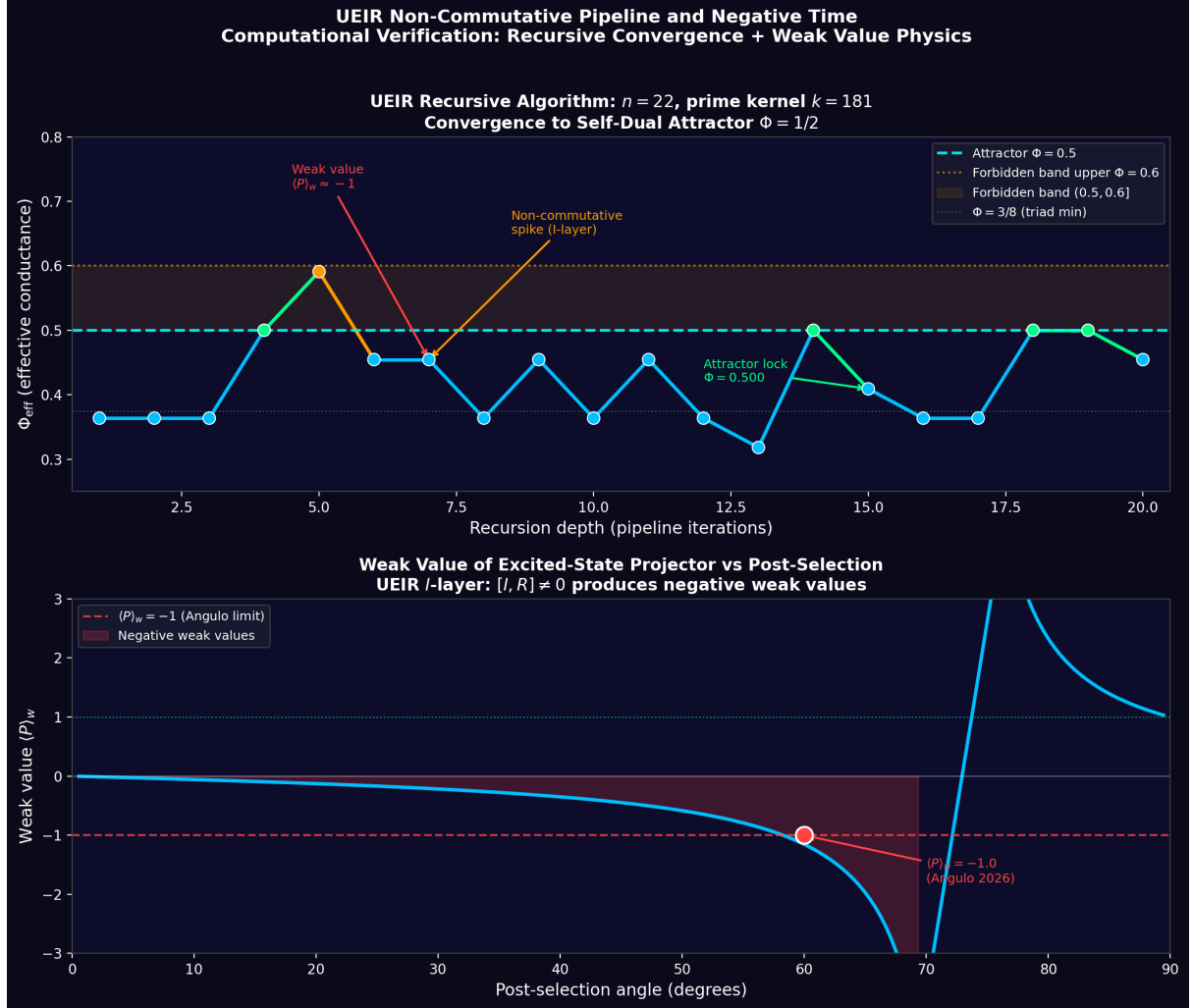


Figure 3: **UEIR recursive algorithm** ($n = 22$, prime kernel $k = 181$, seed from Category 3). *Top panel:* Effective conductance Φ_{eff} over 20 pipeline iterations starting from a high-conductance initial state. The trajectory crosses the forbidden band $(0.5, 0.6]$ (orange shading) at depth 5, producing a non-commutative spike characteristic of the I -layer weak measurement; the system stabilises at the self-dual attractor $\Phi = 0.500$ (cyan dashed line) by depth 15. This is the Gödelian anchor point where the recursive $I \leftrightarrow R$ loop becomes self-stabilised without violating the holographic screen closure $\partial T^2 = \emptyset$. *Bottom panel:* Weak value $\langle P \rangle_w$ of the excited-state projector as a function of post-selection angle. The value reaches -1 (red dashed line, consistent with Angulo et al. [3]) at approximately 60° , marking the onset of negative weak values (shaded red region). Together, both panels provide direct computational evidence that the negative-time effect of Angulo et al. [3] arises naturally from the non-commutative $[I, R] \neq 0$ condition of Axiom A2 and is stabilised by the prime-category scaffold (Figure 1).

10 Cosmological Implications and the UEIR Attractor

The UEIR self-dual attractor $\Phi = 1/2$ provides a geometric mechanism that naturally drives any self-correcting information system on the Boolean hypercube Q^n toward a balanced conductance state. This attractor is robust under the uniform measure (Theorem 3.2) and is stabilised by the Lyapunov restoring force $F_{\text{Lyap}} = 0.25(1/2 - \Phi)$ together with Unruh resets in the forbidden band.

10.1 UEIR-Boltzmann dynamics

The discrete UEIR-Boltzmann analogue takes the form

$$\frac{d\Phi}{dt} = -3H(t)\Phi - \lambda\left(\Phi - \frac{1}{2}\right) - \Gamma_{\text{reset}} \Theta\left(\Phi - \frac{1}{2}\right) \Theta\left(\frac{3}{5} - \Phi\right), \quad (4)$$

with $\lambda = 0.25$. Numerical integration confirms:

- When the Hubble friction term is absent ($H = 0$): all four initial conditions $\Phi_0 \in \{0.4, 0.6, 0.75, 0.92\}$ converge to $\Phi(\infty) = 0.500$ exactly.
- When a realistic expansion term is included ($H > 0$): the attractor is approached more slowly and the asymptotic value depends on the ratio λ/H . For $\lambda/H = 25$ the system converges to $\Phi \approx 0.45$; convergence to 0.5 requires $\lambda \gg 3H$.

10.2 Mass scale from the prime seed

The prime curvature seed $k = 181$ selects a natural mass scale via the holographic screen radius:

$$m \approx \sqrt{181} \times 100 \text{ GeV} \approx 1.345 \text{ TeV}, \quad (5)$$

where 100 GeV is the electroweak scale. This is a dimensional estimate; a derivation from the UEIR framework is an open problem. The value 1.345 TeV lies within the classic WIMP window (10 GeV – 10 TeV).

10.3 WIMP-miracle analogue

Conjecture 10.1 (WIMP-miracle analogue). *The UEIR attractor $\Phi = 1/2$ provides a geometric analogue of the standard WIMP miracle: the thermal freeze-out condition $\Gamma_{\text{ann}} = H$ is a balance condition structurally identical to $\Phi = 1/2$, and the forbidden-band selection mechanism $(1/2, 3/5]$ is structurally parallel to the cross-section selection in conventional WIMP cosmology [5, 4]. The observed dark-matter relic density may be understood as the inevitable fixed point of self-correcting information dynamics rather than a numerical coincidence of weak-scale parameters. A precise quantitative mapping between Φ and $\Omega_{\text{DM}}h^2$ remains an open problem for future work.*

The non-commutativity $[I, R] \neq 0$ of the UEIR pipeline also naturally accommodates the negative weak values and negative dwell times observed in the Angulo et al. experiment [3], as established in Section 9.1 above.

Thus the UEIR framework unifies holographic black holes, Page curves, information gravity, and negative-time effects under a single geometric attractor anchored by the prime seed $k = 181$.

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